

GSM BASED MESSAGING Using Arduino

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Today, GSM networks are widely-used, whether for making calls or sending SMSes. But some places need urgent messages like in colleges, railway stations, events, share market and so on, and this information should be in real-time. This project is an experiment to give a start to the era of sending messages in real-time. It is about writing the message, which is to be displayed on mobiles, and sending it as an SMS to the receivers. The received message is fetched into a microcontroller (MCU) and, after authentication, is displayed on an LCD screen.

Circuit and working

Circuit shown in Fig. 1 operates using a 12V power supply (connected across CON1). An Arduino MCU requires only 5V but a GSM modem requires 12V. As Arduino Uno has an inbuilt 5V voltage regulator, a common 12V supply can be used for the whole system.

The brain of the circuit is Arduino Uno MCU board (BOARD1). A 16x2 LCD display (LCD1) is used for receiving and displaying the messages. Pins of LCD1, namely, RS, EN, K,

D4, D5, D6 and D7 are connected to Arduino digital pins 12, 11, 5, 4, 3 and 2. When you want to show some information or message, you send an SMS to the GSM modem. Then, Arduino's MCU reads the GSM modem and sends it to the LCD. Here, the LCD is used in 4-bit mode,

which means only four data lines are required to display the data.

GSM modem SIM900A (connected with CON2 and CON3) sends the commands in text mode to Arduino Uno using an RS232 interface. CON3 is connected with DB9 connector of the GSM modem. RS232 voltage

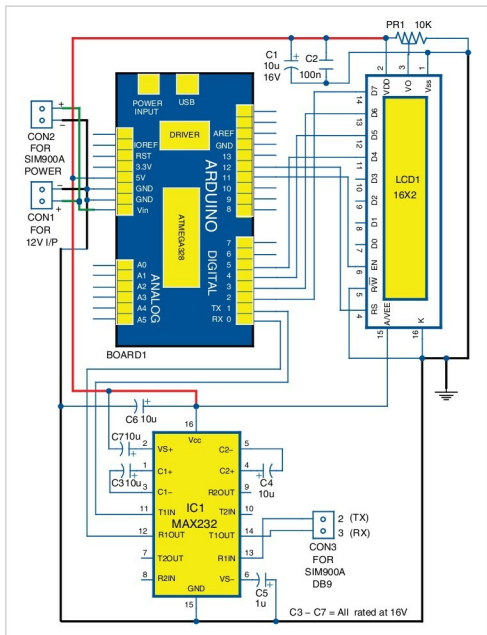


Fig. 1: Circuit diagram of the GSM based messaging

PARTS LIST

Semiconductors:

- BOARD1 - Arduino Uno
- IC1 - MAX232

Resistors (1/4-watt, 5%):

- PR1 - 10-kilo-ohm preset

Capacitors:

- C1, C3, C4, C6, C7 - 10µF, 16V electrolytic
- C2 - 100nF, ceramic disk
- C5 - 1µF, 16V electrolytic

Miscellaneous:

- LCD1 - 16x2 alphanumeric display
- CON1-CON3 - 2-pin connector
- GSM modem SIM900A
- DC 12V/12V battery